

Shreyas Deb

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EDUCATION

International Institute of Information Technology, Hyderabad

Aug 2023 – May 2028

B.Tech. in Computer Science & M.S. by Research in Computational Natural Sciences

Hyderabad, Telangana

- **CGPA:** 8.85 / 10.0
- **Achievements:** Dean's List (Top 15%: Monsoon 2024), Merit List (Top 20%: Monsoon 2023).
- **Relevant coursework:** Data Structures & Algorithms, Operating Systems & Networks, Computer Systems Organization, Distributed Systems, Algorithm Analysis and Design, Design & Analysis of Software Systems, Statistical Methods in AI, Probability & Random Processes, Pricing Options with Mathematical Models (Coursera).

EXPERIENCE

Undergraduate Researcher

Aug. 2025 – Present

Center for Computational Natural Sciences and Bioinformatics (CCNSB), IIIT Hyderabad

Hyderabad, Telangana

- Conducting research under **Dr. Vinod P K**, focusing on computational modeling of complex biological systems through the lens of **Systems and Network Biology**.
- Investigating the intersection of **Deep Learning** and **Network Biology** to decipher emergent disease mechanisms, integrate multi-omics data, and predict clinical outcomes for **precision medicine**.
- Built and evaluated **deep learning architectures** for high-dimensional **multi-omics data integration**, emphasising model design, experimental rigour, and interpretable prediction of clinical outcomes.
- Reimplemented **5+ ML/DL pipelines** for noisy, heterogeneous biological data, focusing on quantitative modelling and representation learning.

Software Development Intern

Jun. 2026 – Jul. 2026

CloudNuro.ai

Hyderabad, Telangana

- Contributed to multi-tenant **LLM Inference Gateway** enabling secure BYOK deployments across multiple LLM providers.
- Built observability pipelines with **OpenTelemetry** and **Langfuse** for distributed tracing, latency monitoring, and inference analytics.
- Built a **LangGraph-based multi-agent software development suite** that ingested sprint transcripts/documents and repositories to automate **requirements extraction, ticket generation, UX generation, and code generation**.
- Designed **human-in-the-loop self-refinement loops** at each agent phase, enabling iterative validation and modification of requirements, tickets, UX frames, and generated code until satisfactory results were achieved.
- Integrated configurable **design systems and reusable skills** to generate responsive desktop/mobile UX frames, achieving **70–80% visual alignment** with the company's design system when evaluated against historical sprints.

AI/ML Engineering Intern

May 2025 – Jul. 2025

MontyCloud

Bengaluru, Karnataka

- Built a production-grade **Agentic Suite** for Product Management as part of the **PM360 team**, including a **Spec Writer**, **Meeting Analyzer**, and **Product Value Agent**, increasing PM efficiency by **40%**.
- Designed **multi-agent LLM systems** using **GraphRAG** pipelines grounded in a **Neo4j Knowledge Graph** with temporal memory and vector search (via **Azure AI Search**), deployed on a fully **serverless AI backend** using **AWS**, with agent workflow orchestration via **AWS Strands** and **MCP (Model Context Protocol)** servers.
- Engineered data ingestion and retrieval pipelines from **Jira** and **Confluence**, and delivered a system deployed across **US and India**, achieving **80%+ reduction** in spec creation time and **90%+ spec accuracy**.

Software Engineering Intern

Jan. 2025 – Apr. 2025

uExcelerate

Hyderabad, Telangana

- Developed a scalable **Learning Management System (LMS)** with an enhanced UI and personalized content delivery, increasing user engagement by **35%** and improving average session duration by **25%**.
- Designed and implemented a modular **recommendation pipeline** using hybrid ranking (content-based, collaborative, and interest-based), achieving **80%+ accuracy** with clean extension points for future personalization features.

Front-end Development Intern

Jan. 2025 – Feb. 2025

Texperia

Bengaluru, Karnataka

- Optimized the frontend architecture for the **Playda platform**, achieving a **32% reduction** in average page load time and improving interaction responsiveness by **~25%** for monthly visitors.
- Contributed to the **U.S. market expansion initiative** by redesigning and localizing U.S.-facing pages, directly supporting a projected **40% increase** in user acquisition at launch.

- Natsuki – Hybrid Search Engine with Learned Ranking** | *Python, Information Retrieval, LightGBM* [GitHub](#)
- Built a **hybrid search engine from scratch** (inverted-index **BM25**, dense retrieval, reciprocal rank fusion), validated against real relevance judgments across **3 BEIR datasets** spanning scientific claims, medical QA, and argument retrieval.
 - Trained a **LambdaMART learned reranker** (LightGBM) on 138K labeled query-document candidates, improving **NDCG@10 from 0.67 to 0.75** by learning retrieval-signal weights from data instead of a fixed fusion formula.
 - Implemented **KD-tree and LSH** nearest-neighbor search **from scratch**, empirically confirming the **curse of dimensionality** (KD-tree touches 100% of nodes at 384-dim) and demonstrating a tunable **recall/latency tradeoff** with LSH.
 - Integrated **LLM query expansion** via the **Mistral API**, **A/B testing** it against a **NDCG/MRR/Recall@k** evaluation suite backed by 54 unit tests.
- StratBacktester – Strategy Backtesting Framework** | *Python, Quantitative Finance* [GitHub](#)
- Built a modular **Python backtesting framework** to evaluate systematic trading strategies on historical market data, separating data ingestion, signal generation, portfolio accounting, and performance analysis.
 - Implemented configurable execution and risk controls, including transaction-cost-aware portfolio simulation, position tracking, and reproducible experiment workflows for strategy research.
 - Generated quantitative evaluation **reports covering return, volatility, drawdown, win rate, and risk-adjusted performance metrics** to compare strategies against baselines.
- PairsTrader – Statistical Arbitrage Engine** | *Python, Mathematical Modelling, Statistics, Finance* [GitHub](#)
- Built a modular statistical-arbitrage engine for NSE equities, using **Engle-Granger cointegration** testing and **PCA + DBSCAN clustering** to screen a 30+ stock universe down to statistically mean-reverting pairs.
 - Implemented both rolling-window **OLS** and online **Kalman-filter** hedge-ratio estimation (time-varying beta/alpha via recursive Bayesian updating vs. discrete window re-fitting), letting the spread model adapt to regime shifts without relying on a single full-sample fit.
 - Engineered 13 mean-reversion features (AR(1) half-life, momentum, volatility, rolling correlation) and trained an **XGBoost** classifier to gate entries by predicted reversion probability, evaluated against a rule-based z-score baseline in a full bar-by-bar backtester with transaction costs, Sharpe ratio, and max drawdown.
- StockSeqCompare – Sequence Model Benchmark for Stock Return Prediction** | *Python, PyTorch* [GitHub](#)
- Built and benchmarked **6 sequence architectures** (LSTM, GRU, Transformer, BiLSTM, LSTM+Attention, CNN-LSTM) predicting next-day log returns on 5 years of AAPL data; best model (LSTM) reached **54.7% directional accuracy** and an annualized **Sharpe ratio of 1.15** on a held-out test set.
 - Engineered **10 causal technical indicators** (MACD, RSI, rolling volatility, momentum, Bollinger Bands) with expanding-window z-score normalization, verified via unit tests to prevent look-ahead bias in the non-stationary price series.
 - Backtested a long/short strategy per model against a buy-and-hold benchmark (+15.3% over the window); **2 of 6** architectures beat buy-and-hold, with the best (LSTM) generating **+22.1% total return** at **-19.8%** max drawdown.
- Cuffka – Distributed Message Broker** | *Go, Raft Consensus, Distributed Systems, TCP, MessagePack* [GitHub](#)
- Built a **Apache Kafka reimplementation** in **Go** featuring a **Raft-based controller quorum**, append-only partition logs with in-memory offset indexing, consumer group coordination, and a custom **MessagePack** wire protocol over TCP.
 - Implemented a complete **Raft consensus engine** from scratch – leader election, log replication with **fast-backtrack conflict resolution**, and snapshotting with **async disk I/O**, full crash recovery from snapshot and log delta.
 - Engineered **generation-based consumer group rebalancing** with heartbeat-driven eviction, **durable offset commits** via atomic rename, and broker failure detection with **partition leader reassignment**
- Docs++ – Distributed File System** | *C, Networking, System Design* [GitHub](#)
- Built a distributed document system from scratch in **C** with **sentence-level locking** for concurrent multi-user editing, real-time access control, and persistent versioned storage.
 - Architected a **3-tier distributed system** with a Name Server, multiple Storage Servers, and clients, supporting **12+ filesystem operations** including streaming, hierarchical namespaces, versioning, and ACL-based permissions.
 - Implemented checkpoint-based version control, undo support, persistence, and efficient file lookup with **zero external libraries**.
- AIshA – Intelligent Unix Shell** | *C, OpenSSL, Systems Programming* [GitHub](#)
- Built a **POSIX-compliant Unix shell** in **C** from scratch, implementing **40+ built-in commands**, multi-stage command pipelines, and full **I/O redirection** (input, output, append, and error streams).
 - Designed a modular **command execution engine** using **fork/execve** process management, with **job control** (foreground/background execution, **fg/bg/jobs**) and **asynchronous signal handling** (**SIGCHLD, SIGINT, SIGTSTP**) for correct POSIX process lifecycle semantics.
 - Implemented **termios-based raw-mode line editing** with persistent command history, tab-based autocomplete, and cursor/line navigation, replicating core readline behavior without external libraries.
 - Integrated **natural-language command execution** by building a custom **HTTPS client** (over raw sockets with **OpenSSL**) to query **Gemini AI**, including a hand-written **JSON parser** to deserialize model responses into executable shell commands.

- Saathi – AI Mental Wellness Platform | *Next.js, FastAPI, LangChain/LangGraph, PostgreSQL* [GitHub](#)
 - Designed a **LangGraph-based multi-agent architecture** routing conversations to **six mood-specialist agents**, augmented with **RAG** and safety validation for context-aware, grounded responses.
 - Built a **provider-agnostic model layer**, enabling integration and switching across multiple foundation model providers.
 - Implemented a **retrieval layer** supporting both **pgvector** search and SQLite vector retrieval with hybrid ranking.
 - Strengthened platform security by ensuring no **cross-user data exposure**, introducing **anonymous posting**, and implementing **server-side access control** and **rate limiting** for authentication endpoints.
- shAI – Natural Language Shell Assistant | *Python, AI Agents* [GitHub](#)
 - Built a **framework-free CLI** that translates natural language into executable shell commands, with a pluggable provider layer supporting **local (Ollama)** and **hosted (Mistral, OpenAI-compatible)** LLM backends.
 - Designed a **multi-agent validation pipeline** separating query intent classification, command generation, and safety analysis to block destructive or malicious commands before execution.
- TradeHub – Campus E-Commerce Marketplace | *React, Node.js, Express, MongoDB, Cloudinary* [GitHub](#)
 - Built a full-stack **MERN** campus marketplace for IIIT Hyderabad, enabling peer-to-peer buying and selling with **JWT-based authentication**, **bcrypt** password hashing, and **Cloudinary**-backed image management.
 - Implemented optimistic UI updates via **Zustand** state management and added a **Gemini 1.5 Flash** enabled chatbot
 - Designed a modular **REST API** with **Express** and **Mongoose**, handling **multipart/form-data** uploads via **Multer** and enforcing **CORS** policies across a decoupled frontend-backend deployment.

ACHIEVEMENTS

- Atlassian FastTrack 2026** – Selected as a winner among **300+ eligible candidates** and received PPI offers.
- HackIIIT 2025 Most Impactful** – Won Most Impactful award at HackIIIT 2025 for Saathi, an AI mental wellness platform.

TECHNICAL SKILLS

Languages: C, C++, Go, Rust, Python, JavaScript/TypeScript, SQL, Bash
Frameworks & Libraries: FastAPI, Flask, React, Next.js, Node.js, Express.js, PyTorch, TensorFlow
Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Neo4j, DynamoDB, Vector Databases
Cloud & Infrastructure: Linux, Docker, Kubernetes, Microservices Architecture (AWS, LocalStack), Git
Concepts: System Design, Distributed Systems, Computer Networks, Deep Learning & Machine Learning

LEADERSHIP & EXTRACURRICULAR

- Tech Team Head

Aug. 2025 – Apr. 2026

E-Cell IIIT-Hyderabad

 - Directed all technical operations for the Entrepreneurship Cell, maintaining server infrastructure at **98% uptime**.
 - Led the engineering team for **Megathon 2025**, Hyderabad’s largest student-run hackathon (**1,000+ participants**), overseeing registration portals and live event dashboards.
 - Assisted in organizing flagship events and workshops, fostering a spirit of entrepreneurship and innovation across the campus community.
- Project Manager & Maintainer

Mar. 2025 – Apr. 2026

Open Source Developers Group (OSDG), IIIT-Hyderabad

 - Maintain **Whispr**, a privacy-first open-source review platform serving **1,000+ students** across **200+ courses** and **100+ professors**.
 - Mentor open-source contributors, run technical workshops, and steward student-run repositories to grow the campus open-source ecosystem.
 - Drive contribution culture by onboarding new developers, establishing code review standards, and organizing contributor sprints.